# WEYRO

Automotive Marketplace Platform — Production Feature & Architecture Specification

Production Ready

Generated: October 2026

## 1. Core Platform & Backend Capabilities

### 🔐 Identity, Auth & RBAC

- **Dual-Identifier Sign-in**: Email and Indian Mobile (+91).
- **Password Security**: Bcrypt (cost 12) + constant-time dummy verification for non-existent accounts.
- **Token Pair Lifecycle**: 15-min JWT access token + 7-day stateful rotating refresh tokens in DB.
- **Role Guards**: Global RBAC covering CUSTOMER, PROVIDER, BOTH, and ADMIN.
- **Customer PII Masking**: Contact info withheld from public feeds until quote acceptance.

### 📍 PostGIS Geospatial Engine

- **Native Spatial Point Type**: geography(Point, 4326) synced automatically via DB triggers.
- **Sub-Millisecond Radius Search**: Indexed with PostgreSQL GiST spatial indexes.
- **Spatial Filtering**: ST_DWithin finds workshops within customer-selected radius (e.g., 10-50 km).
- **Cursor Pagination**: High-performance cursor pagination (max limit 50).

### 🛠️ Damage Requests & Bidding

- **Structured Lead Creation**: Multi-angle photos, vehicle details, fuel type, and damage description.
- **Workshop Competitive Bids**: Itemized quotes with labor charges, spare parts, and estimated days.
- **Atomic Quote Acceptance**: Guarded PostgreSQL transactions prevent concurrent double-booking.
- **Scheduled Expiration**: Automated cron jobs (@nestjs/schedule) auto-expire stale requests.

### 💳 Payments & Escrow (Razorpay India)

- **Indian Currency Standard**: Native INR (₹) processing in paise.
- **UPI & Cards**: Instant UPI intent, netbanking, and domestic debit/credit cards.
- **RBI PA/PG Compliance**: Authorized capture architecture without illegal custodial accounts.
- **Webhook Idempotency**: Verified with timing-safe HMAC SHA-256; deduplicated via WebhookEvent.

### 💬 Real-Time Chat & Push Notifications

- **WebSocket Gateway**: Socket.IO on /chat with JWT handshake authentication.
- **Isolated Chat Rooms**: Room isolation (thread_<id>) restricted to authenticated participants.
- **Expo Push Dispatch**: Immediate device push alerts on new quotes, acceptances, and chat messages.
- **Automated Token Pruning**: Purges invalid or unregistered device tokens automatically.

### 📊 Admin & Observability

- **Health Probes**: /api/v1/health (liveness) & /api/v1/ready (readiness with DB query).
- **Request Tracing**: Correlation IDs attached to all HTTP requests via x-request-id.
- **OpenAPI / Swagger**: Interactive API specifications automatically documented at /api/docs.
- **Automated CI**: GitHub Actions CI pipeline running lint, typecheck, tests, and Expo Doctor.

## 2. Supabase Storage Bucket Specifications ( weyro-uploads )

### 📦 Storage Infrastructure Overview

The **weyro-uploads** bucket provides S3-compatible, high-performance object storage hosted on Supabase in AWS Mumbai (ap-south-1) and accelerated globally via Cloudflare CDN.

**Region:** ap-south-1 (Mumbai)

**File Limit:** 10 MB per item

**MIME Whitelist:** JPEG, PNG, WebP

**Architecture:** Presigned S3 PUT URLs

| Bucket Directory | Purpose & Content | Security & Access Policies |
|---|---|---|
| damage-requests/ | Collision, dent, and paint damage photos uploaded by vehicle owners when initiating repair quotes. | Presigned PUT upload (5-min validity). Public CDN read for verified repair workshops. |
| quotations/ | Repair diagrams, inspection estimates, and spare parts invoices submitted by repair workshops. | Restricted upload to verified repair providers. Read access granted to request owner. |
| completion-proofs/ | Before-and-after vehicle repair photos uploaded by workshops upon completing repair jobs. | Uploaded upon booking status transition to IN_PROGRESS or COMPLETED. |
| chat-attachments/ | Media and parts photos shared during live direct messaging between customer and workshop. | Accessible only through thread room authorization. CDN accelerated. |
| provider-profiles/ | Garage storefront photography, workshop logos, trade certifications, and facility photos. | Uploaded during provider registration and profile updates. Public read CDN. |

## 3. Mobile Client Architecture (Expo SDK 57)

### 📱 Hardware Secure Storage

Replaced unencrypted storage with expo-secure-store. Tokens reside strictly in Apple Keychain (iOS) and Android KeyStore with AES-256 encryption.

### 🔄 Resilient Offline Mutation Queue

Network loss triggers an offline mutation queue with unique Idempotency-Key headers and exponential backoff retry algorithms.